

AYUSH NATHANI

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EDUCATION

Indian Institute of Technology (IIT) Patna & Indian Institute of Information Technology (IIIT) Ranchi

Master of Computer Applications (MCA) | Current *CGPA: 10.0/10.0*

August 2025 – Present (Expected Graduation: July 2027)

Amity University Jharkhand

Bachelor of Computer Applications (BCA) | *CGPA: 8.72*

July 2021 – June 2024

SKILLS

Languages: Python, JavaScript, TypeScript

Frameworks: FastAPI, React, Next.js, Express.js

Platforms: MongoDB Atlas, Vercel, Google Cloud, Railway, Microsoft Azure, Render, Netlify, Firebase

Tools: Git, GitHub, Docker

PROJECTS

TurfGrid AI | Autonomous Smart City Command Center [Live Demo] | [GitHub]

Tech Stack: Python, FastAPI, Next.js, MongoDB Atlas Vector Search, Gemini 2.5 Flash, Groq Llama-3.3-70b, FastMCP, Google Maps API

- Architected a **5-agent autonomous platform** managing fan travel, venue operations, and crowd control across 23 FIFA World Cup 2026 venues using **multi-agent orchestration**.
- Implemented **MongoDB Atlas \$vectorSearch** with **768-dim Gemini embeddings** for semantic venue discovery, and engineered **LLM failover from Gemini 2.5 Flash to Groq Llama-3.3-70b** on resource exhaustion, preserving full tool-calling parity and multi-turn context.

ReelForge AI | Autonomous AI Video Generation Platform [Live Demo] | [GitHub]

Tech Stack: Next.js 16, React 19, TypeScript, RunwayML, ElevenLabs, FFmpeg, Server-Sent Events (SSE)

- Architected a **6-agent content pipeline** (Script Writer -> Scene Director -> Video Producer -> Audio Producer -> Viral Optimizer -> Renderer) converting text prompts into production-ready short-form videos.
- Integrated **RunwayML** and **ElevenLabs** into a **serverless FFmpeg rendering pipeline** supporting Text-to-Image, Image-to-Video, Text-to-Speech, and AI sound effect workflows with automated MP4 export, narration sync, audio normalization, and **real-time SSE progress monitoring**.

Email Triage AI | Autonomous Multi-Agent Email Management System [Live Demo] | [GitHub]

Tech Stack: Python, FastAPI, Groq (Llama 3.1-8b-instant), Gmail API, OAuth2, Docker, Pydantic

- Built a production-grade **multi-agent pipeline** (Classifier -> Responder -> Router) autonomously classifying, prioritizing, and drafting tone-adaptive responses for **live Gmail inboxes using Groq LLM inference**.
- Implemented an **automated PII redaction layer** before LLM inference with confidence-based scoring and explainable chain-of-thought reasoning.
- Containerized on **Hugging Face Spaces** via **Docker** with a **real-time dashboard tracking agent** decisions, routing paths, and confidence metrics.

CERTIFICATIONS

Microsoft Azure AI Essentials: Workloads and Machine Learning on Azure

LinkedIn Learning

Completed August 2025

(Introduction to Azure AI workloads and basic machine learning concepts)

Gemini for Google Workspace

Simplilearn Skillup

Completed August 2025

(Practical application of Gemini AI tools for Google Workspace)